

# ABHISHEK BHARDWAJ

Senior Data Scientist & AI/ML Engineering Lead

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## PROFESSIONAL SUMMARY

Results-driven Senior Data Scientist and AI/ML Engineering Leader with 9+ years of experience building end-to-end machine learning solutions across healthcare, life sciences, and insurance domains. Currently leading AI initiatives at Chubb, specializing in insurance fraud detection using RAG (Retrieval-Augmented Generation), LLMs, and Agentic AI workflows. Proven track record of deploying production-grade ML systems that deliver measurable business impact.

Expertise in MLOps, scalable data pipelines, deep learning (PyTorch, TensorFlow), NLP (BERT, GPT, Transformers). Adept at leading cross-functional data science teams and translating complex ML solutions into actionable business strategies.

## CORE COMPETENCIES & TECHNICAL SKILLS

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|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI/ML & Deep Learning       | PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, Decision Trees, Gradient Boosting, Neural Networks (CNN/RNN), Reinforcement Learning, Ensemble Methods |
| Generative AI & LLMs        | GPT-4, LLM Fine-Tuning, RAG (Retrieval-Augmented Generation), Prompt Engineering, LangChain, Vector Databases, Agentic AI Workflows, Multimodal AI                |
| NLP & Information Retrieval | BERT, Transformers, TF-IDF, Cosine Similarity, Named Entity Recognition, Text Classification, Sentiment Analysis, Information Extraction                          |
| MLOps & Deployment          | MLflow, Kubeflow, Docker, Kubernetes, CI/CD Pipelines (GitHub Actions), Model Monitoring, Experiment Tracking, A/B Testing, Model Versioning                      |
| Data Engineering & Big Data | PySpark, Apache Spark, Hadoop, Hive, HDFS, Airflow, Databricks, ETL Pipelines, Data Warehousing, Feature Engineering                                              |
| Cloud & Infrastructure      | GCP (Vertex AI, Dataproc, Dataflow, BigQuery, Cloud Functions), AWS (Lambda, SageMaker), Azure ML, Serverless Architectures                                       |
| Programming & Databases     | Python, PySpark, SQL, HiveQL, Unix Shell Scripting, R   MySQL, BigQuery, SQL Server, DB2, Neo4j (Graph DB)                                                        |
| Analytics & Visualization   | Power BI, Streamlit, Matplotlib, Seaborn, Jupyter Notebooks   Survival Analysis, Bayesian Optimization, Time Series Analysis                                      |
| Leadership & Domain         | Team Leadership (4-5 members), Stakeholder Management, Cross-functional Collaboration   Healthcare, Life Sciences, Insurance, Pharma, Consumer Growth             |

## PROFESSIONAL EXPERIENCE

### Senior Data Scientist II — Data and Applied AI

Chubb | Bengaluru, India | Aug 2024 – Present

**Awards:** Q1 2025 STAR Award • Q3 2025 North America Analytics Team Recognition (Underwriter Claims Summarization) • Q3 2025 Advanced Analytics Spot Award

#### Insurance Fraud Detection & Risk Modeling (RAG + LLMs)

- Architected and deployed scalable ML-based fraud detection system using RAG (Retrieval-Augmented Generation) and Large Language Models (LLMs) to identify potentially fraudulent insurance claims across long-tail claim lifecycles.

- Developed advanced Information Extraction pipelines from unstructured claims data using NLP transformers and BERT-based models, significantly enhancing fraud detection accuracy and early-stage identification capabilities.
- Implemented batch and real-time processing frameworks for insurance risk scoring, enabling proactive fraud intervention during the critical initial claim maturation period.

#### **Underwriters Assistance Tool — AI-Powered Automation**

- Built Coverage Match and Claims Classification systems using advanced Information Retrieval and semantic search frameworks, automating time-consuming manual underwriting processes.
- Developed Claims Summarization tool leveraging LLMs and extractive/abstractive summarization techniques, freeing underwriter bandwidth for strategic business deliveries.

#### **Innovation & POC Projects**

- Agentic BI Tool:
- Designed and built an Agentic AI-powered Business Intelligence tool using LangChain and autonomous agent workflows to enable natural-language querying for data analysts — reducing report generation time by automating complex analytical pipelines.
- **Claims Crosswalk Initiative:** Built a Claims Crosswalk tool using advanced NLP and entity resolution techniques to intelligently link and reconcile data from multiple external sources, accelerating data engineering workflows for internal analytics projects.

### **Decision Science & Engineering Manager**

**Axtria — Ingenious Insights** | Bengaluru, India | Sep 2022 – Jun 2024

#### **Advanced Marketing Mix Modeling (MMM) & Revenue Optimization — Pharma (J&J, Stemline)**

- Developed sophisticated Marketing Mix Models for Immunology and Neuroscience therapeutic areas using ensemble methods (Random Forest, XGBoost) and time series analysis, achieving ~10% increased revenue through optimized marketing spend allocation.
- Implemented Bayesian optimization for hyperparameter tuning, improving model accuracy by 15%; leveraged PySpark on Databricks for distributed processing of large-scale marketing datasets.
- Designed custom genetic algorithm for multi-objective optimization of marketing budget allocation across omnichannel touchpoints.

#### **Omnichannel Marketing Attribution & Customer Journey Analytics**

- Designed and deployed a multi-touch attribution model using Markov Chains and SHAP values for model interpretability, tracking customer journeys across digital and traditional channels.
- Built end-to-end MLOps pipeline using MLflow for experiment tracking and model versioning; processed large-scale customer interaction data on GCP BigQuery and Databricks.

#### **Pharmaceutical Rep Communication System (POC) — GenAI**

- Architected end-to-end AI system generating personalized pharma rep-to-doctor communications using GPT-4, integrating knowledge graph (Neo4j), recommendation engine, and NLP generation.
- Deployed to production using Flask + Streamlit with CI/CD pipeline in GitHub Actions; orchestrated ML workflows using Kubeflow with MLflow experiment tracking.

### **Data Science & ML Engineering Consultant**

**EXL (Contracted to CVS Health / Aetna)** | Noida, India | Jul 2016 – Sep 2022

**Career Progression:** Business Analyst → Senior Business Analyst → Assistant Manager → Manager, Data Science & ML Engineering

**Awards:** Led DS Offshore Team with 5/5 SLA ratings for 3 consecutive quarters • 2019 Q3 STAR Award (Underwriting Automation) • 2017 Outstanding Team Award • 2016 Promising Newcomer Award

#### **Customer Lifetime Value (CLV) Prediction at Scale**

- Developed distributed Random Forest model using PySpark to predict CLV for 2M+ prospects; implemented feature engineering pipelines using Spark SQL for efficient large-scale customer data processing.
- Utilized GCP Dataproc for orchestrating scalable model training and scoring, reducing processing time by 70%.

#### **Patient Readmission Risk Prediction — Deep Learning**

- Built production ML pipeline for readmission risk prediction incorporating clinical notes; leveraged PyTorch with BERT for rich text feature extraction from unstructured clinical data, improving model AUC from 0.82 to 0.89.

- Implemented end-to-end MLOps workflow: Airflow for data ingestion, PySpark for distributed processing, MLflow for experiment tracking, and automated model monitoring for drift detection.

#### ***NLP-based Plan Sponsor Entity Matching***

- Designed scalable string matching algorithm using TF-IDF and cosine similarity on Spark MLlib, achieving 75% accuracy matching 20K plan sponsors across 40K company name records — saving hundreds of hours of manual work.

#### ***Survival Analysis & Predictive Analytics***

- Implemented Kaplan-Meier estimator and Cox proportional hazards model for patient outcomes and customer attrition analysis; developed custom feature importance method combining SHAP values and permutation importance.
- Created interactive Streamlit application for healthcare providers to visualize survival curves and risk factors, driving 10% improvement in patient outcomes through data-driven decision-making.
- Designed and deployed automated marketing optimization strategies using advanced propensity modeling and A/B testing, enhancing customer acquisition across omnichannel touchpoints.

## **EDUCATION**

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### **Bachelor of Technology (B.Tech.) — Mechanical Engineering**

Motilal Nehru National Institute of Technology (MNNIT), Allahabad | 2012 – 2016

*Minor: Computational Fluid Mechanics*

## **CERTIFICATIONS**

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- Deep Learning Specialization — deeplearning.ai (Coursera)
  - Google Cloud Platform (GCP) Practitioner — Data Science & Engineering Pipelines
  - Machine Learning: Algorithms in the Real World Specialization
  - CFA Level 1 Candidate

## **AWARDS & RECOGNITION**

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- Chubb Q1 2025 II STAR Award — Excellence in insurance fraud and risk modeling
  - Chubb Q3 2025 North America Analytics Team Recognition — Underwriter Claims Summarization
  - Chubb Q3 2025 Advanced Analytics Spot Award
  - EXL 2022 — Led DS Offshore Team (3 FTEs) with consistent 5/5 SLA rating across 3 consecutive quarters
  - EXL Q3 2019 STAR Award — Underwriting Automation
  - EXL 2017 — Outstanding Team (NAB QA & DE)
  - EXL 2016 — Promising Newcomer Award